



MEDFIND

Offline Hospital Locator



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PROJECT OVERVIEW:

Field	Detail
Project Name	MedFind: Offline Hospital Locator
Project Type	Mobile Application (Android and iOS)
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MedFind is a mobile application that helps users locate nearby hospitals and clinics in emergency situations without requiring an active internet connection. It uses the device's GPS combined with a pre-downloaded local database of medical facilities to provide instant, offline access to critical location data.

PROBLEM STATEMENT:

In emergency situations — accidents, sudden illness, rural travel — people often cannot find nearby hospitals quickly due to poor connectivity or unfamiliarity with the area. Existing solutions like Google Maps require an active internet connection, making them unreliable in low-network zones.

- No offline-capable hospital locator exists for Indian rural/semi-urban regions.
- Bystanders at accident scenes waste critical time searching for hospitals.
- Rural users frequently have no data connectivity when medical emergencies occur.
- Emergency responders lack a fast, structured tool to identify nearest facilities.

PROJECT OBJECTIVES

- Provide offline hospital and clinic discovery using device GPS.
- Maintain a locally stored, updatable database of medical facilities.
- Display distance and direction to nearest hospitals without internet.
- Support emergency responders, general public, and rural users equally.
- Enable one-tap calling directly from the app.

USER ROLES AND LOGIN MODULES

- Single user role, no login required.
- The app is open access by design.
- In an emergency, a login screen is a barrier that costs lives.

PROJECT MODULES

Module	Description
GPS MODULE	Detect user's current location using device's GPS
OFFLINE DATABASE MODULE	Pre-download SQLite database of hospitals per city
SEARCH AND FILTER MODULE	Filter by type (govt./pvt), distance, specialty
OFFLINE MAP MODULE	City map downloaded by user before use
CALL MODULE	One-tap call to selected hospital directly from app
MAP DOWNLOAD MANAGER	Download/manage city maps; detect when user exits download zone
OUT-OF-ZONE ALERT MODULE	Notify user when GPS detects the user has moved outside their downloaded city.
HOSPITAL REPORT MODULE	Allows users to submit missing or incorrect hospital details via in-app form

FUNCTIONAL REQUIREMENTS

- Detect user location via GPS without internet.
- Display hospitals sorted by distance from current location.
- Show hospital name, address, contact number, type (govt/private), and specialty.
- Allow filtering by distance and hospital type.
- Render offline map using pre-downloaded city map data.
- Enable one-tap calling to hospital from the listing.
- Allow user to download maps for specific cities manually.
- Prompt user to download at least one city map on first launch before proceeding.
- Detect when user's GPS location exits the downloaded city boundary.
- Send a notification prompting the user to download the new city's map.
- Allow downloading updated hospital database per city when internet is available.
- User can submit a report for a missing or incorrect hospital via in-app form.
- Report is queued locally if submitted offline and sent automatically when internet is available.
- Submitted reports go to admin review before being added to the database.

NON-FUNCTIONAL REQUIREMENTS

- App must function fully offline after city map is downloaded.
- GPS fetch must complete within 5 seconds.
- Out-of-zone detection must trigger notification within 2 minutes of boundary exit.
- App size must stay under 50MB excluding downloaded map data.
- Map downloads should be manageable — user can delete old city maps to free storage.
- UI must be usable under panic/emergency conditions — large text, minimal steps.
- Must support Android 8.0+ and iOS 13+.

SYSTEM FLOW

- **Normal Use:** User Opens App → GPS Fetches Location → Query Local SQLite DB → Display Nearby Hospitals → User Selects Hospital → One-Tap Call
- **First Launch:** App Opens → Prompts User to Download City Map → User Downloads → App Ready to Use
- **Out-of-Zone:** GPS Detects Location Outside Downloaded City → App Sends Notification → User Downloads New City Map
- **Database Update (Online):** App Detects Internet → Checks for DB Update → Downloads Latest Data → Stores Locally

TECHNOLOGY STACK

LAYER	TECHNOLOGY
FRONTEND / APP	Flutter (Android + iOS)
LOCAL DATABASE	SQLite (via sqflite package)
GPS	Flutter Geolocator package
OFFLINE MAPS	Flutter Map + OpenStreetMap tiles (cached)
BACKEND (DB UPDATES)	Java Spring Boot
CLOUD DATABASE	MySQL (for admin-managed master data)
HOSTING	AWS / Firebase

CONCLUSION

MedFind addresses a real, life-critical gap in India's emergency response ecosystem. By combining offline-first architecture with GPS and a structured hospital database, it delivers reliable medical facility discovery exactly when internet access fails. The hospital database is maintained by the system administrator. The project demonstrates practical application of mobile development, database management, and emergency UX design.